

Geometry of Recursive Knots:

Iterated Cables by Frame Offsets, Their Thickness and Ropelength

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Abstract

We study a concrete family of *recursive knots*: starting from a round circle K_0 , K_d is the (p_d, q_d) torus pattern on the tube of radius r_d around K_{d-1} , placed with a rotation-minimising (Bishop) frame closed by a linear twist, and ask how much rope (length over thickness) it needs. We prove that K_d is embedded and smooth when $r_d < \text{thick}(K_{d-1})$; that K_d is an explicit cable of K_{d-1} (the closed frame has linking number $\text{Wr } K_{d-1} - \alpha_d/2\pi$); two-sided length bounds; and, for $p_d = 2$, $\text{thick}(K_d) \leq r_d$, so $\text{Rop}(K_d) \geq L(K_d)/r_d$. For the lower bound $\text{thick}(K_d) \geq \min(\text{thick } K_{d-1} - r_d, r_d/2)$ with $p_d = 2$, $r_d \leq \text{thick}(K_{d-1})/2$: in the default family (2,3) every doubly critical pair of K_d with distinct base points is at distance $\geq \min(2(\text{thick } K_{d-1} - r_d), r_d)$ at every depth, so the bound reduces to $\text{minRad}(K_d) \geq r_d/2$, which we prove at depth one by hand and at depth two by interval arithmetic on the explicit base K_1 ($\text{minRad}(K_2) \geq 0.842 r_2$ for every $f \leq 1/2$ and all phases). For $d \geq 3$ it follows from a bound (H_3) on derivatives of the speed and of the curvature vector of K_{d-1} , and fails for an arbitrary base without one. The speed and curvature of K_d are controlled by the (H_3) data of K_{d-1} , but $\mu(K_d)$, the size of the derivative of the curvature vector, cannot be bounded by any locally bounded function of the (H_3) data of an arbitrary base (in the family μ grows slowly for the computed levels); (H_3) uniformly in d remains open. Under the thickness bound (proved for $d \leq 2$), $\text{Rop}_d \leq 2\pi\Lambda_{\text{Rop}}^d$ with an explicit Λ_{Rop} . Explicit polygons with up to 8192 vertices give $L/\tau = 38.4, 155.6, 622.7$ at depths 1, 2, 3 for (2,3) and $f = 1/2$, stable to 0.03% under doubling of the resolution: numerically verified upper bounds for explicit polygons, not certified smooth thicknesses. No stationarity, optimality or recursive flow law is claimed.

1 Introduction and status

The research line. Line RTK001 (“Geometry of recursive knots”) asks how nested knots—knots tied on the tube of other knots, recursively—can be represented as tubes without self-intersection, and how much length that costs. Its public record states one finding (a feasible upper bound at depth one) and an explicit scope restriction: neither stationarity nor optimality of any configuration is certified, and no general recursive flow law has been established. No manuscript existed before this draft. Five internal reviews have been applied (v0.2–v0.6, 30/09–03/10/2026); the thickness hypothesis of v0.1 turned out false for an arbitrary base (Remark 3.13).

What is done here. We fix a definite family (Section 2); prove embeddedness, the framing, two-sided length bounds, an upper bound on the thickness and most of the lower bound, isolating what is not proved (Section 3); compute explicit polygons at depths 1 to 3 with a resolution check (Section 4); and tabulate every claim with its status (Section 5). Lower bounds for general knots are only quoted (Remark 3.24).

2 Setting

2.1 Thickness and ropelength

Let $K \subset \mathbb{R}^3$ be a closed $C^{1,1}$ curve of length $L(K)$. Its *thickness* $\text{thick}(K)$ is the supremum of ρ such that the open tube of radius ρ about K is embedded (normal injectivity radius). Two equivalent descriptions

are classical. Litherland, Simon, Durumeric and Rawdon [11] show, for C^2 curves (extended to the $C^{1,1}$ case in [3]; all curves K_d below are C^∞),

$$\text{thick}(K) = \min(\min\text{Rad}(K), \tfrac{1}{2} \text{dcsd}(K)), \quad (1)$$

where $\min\text{Rad}$ is the minimal radius of curvature and dcsd the *doubly critical self distance*, the minimal distance between pairs $x \neq y$ of K whose chord is perpendicular to K at both ends. Gonzalez and Maddocks [7] show $\text{thick}(K) = \inf_{x \neq y} \rho_{pt}(x, y)$, with $\rho_{pt}(x, y) = |x - y| / (2 \sin \angle(T_x, y - x))$ the radius of the circle through y tangent to K at x . The *ropelength* is $\text{Rop}(K) = L(K) / \text{thick}(K)$; it is scale invariant, $\text{Rop}(K) \geq 2\pi$ for every closed curve (Fenchel's theorem [5], total curvature $\geq 2\pi$, together with $\kappa \leq 1/\text{thick}$), with equality only for the circle, and $\text{thick}(K) \leq \min\text{Rad}(K)$ always.

2.2 Frames, holonomy and the closing twist

Let K be a closed C^2 regular curve parametrised on $[0, 2\pi)$ (not necessarily by arclength), with unit tangent T and curvature κ . A *rotation-minimising* or Bishop frame [1] is an orthonormal frame (T, N_0, B_0) with $N'_0 = -\kappa_1 |K'| T$, $B'_0 = -\kappa_2 |K'| T$, $\kappa_1^2 + \kappa_2^2 = \kappa^2$; it is unique up to a constant rotation in the normal plane (fixed in Definition 2.1) and is in general *not* periodic: after one turn the transported normal is rotated by a holonomy angle $\alpha \in (-\pi, \pi]$. We close it by the linear twist $\omega(t) = -\alpha t / 2\pi$, i.e. $N = \cos \omega N_0 + \sin \omega B_0$, $B = T \times N$. The closed frame satisfies

$$N' = -\tilde{\kappa}_1 |K'| T + \omega' B, \quad B' = -\tilde{\kappa}_2 |K'| T - \omega' N, \quad |\omega'| = \frac{|\alpha|}{2\pi} \leq \frac{1}{2}, \quad (2)$$

with the rotated curvature components $\tilde{\kappa}_1 = \kappa_1 \cos \omega + \kappa_2 \sin \omega$, $\tilde{\kappa}_2 = -\kappa_1 \sin \omega + \kappa_2 \cos \omega$ (same norm κ); below we write κ_1, κ_2 for whichever frame is in use.

2.3 The family of recursive knots

Definition 2.1 (iterated cable by frame offsets). Let K_0 be the circle of radius $R_0 = 1$. For $d \geq 1$ choose coprime integers $p_d \geq 2$, $q_d \geq 1$, a radius $r_d > 0$ and a phase φ_d . Reparametrise K_{d-1} to period 2π by the linear change $t \mapsto t/p_{d-1}$ (so that K_{d-1} is traversed at parameter speed $|K'_{d-1}|$, not at unit speed), let (N_{d-1}, B_{d-1}) be its closed Bishop frame, normalised by $N_{d-1}(0) =$ the unit projection of e_z onto the normal plane at $K_{d-1}(0)$ (of e_x if $|T(0) \cdot e_z| > 0.9$), and set for $t \in [0, 2\pi p_d)$

$$K_d(t) = K_{d-1}(t \bmod 2\pi) + r_d [\cos \theta_d(t) N_{d-1}(t) + \sin \theta_d(t) B_{d-1}(t)], \quad \theta_d(t) = \frac{q_d}{p_d} t + \varphi_d. \quad (3)$$

Rotating $N_{d-1}(0)$ by an angle ϑ amounts to replacing φ_d by $\varphi_d + \vartheta$, so a choice of normalisation is a choice of phases; for $p_d = 2$, q_d odd, $\varphi_d + \pi$ gives the same curve. The default is $(p_d, q_d) = (2, 3)$, $\varphi_d = 0$ at every depth; r_d is a fraction f of the thickness of K_{d-1} (the guaranteed lower bound ρ_{d-1} in the theory, the measured polygonal value in the experiments).

K_1 is the (p_1, q_1) torus knot on the round torus of radii $(1, r_1)$; the knot type of K_d for $d \geq 2$ depends on the framing, which Lemma 3.2 makes explicit. All results of Section 3 hold for every choice of $N_{d-1}(0)$ and φ_d ; computed lengths and thicknesses do not depend on it at the printed precision, but the margins of Conjecture 3.25 do (Section 4). The pattern angle advances in the inherited parameter t , not in arclength, so the *pitch* of the strands (h in Section 3) is a property of the family, not a free parameter.

2.4 Polygonal measurements

Numerically a curve is a closed polygon $P = (x_0, \dots, x_{N-1})$. We use $L(P) = \sum |x_{i+1} - x_i|$; $\min\text{Rad}(P)$, the minimum circumradius of consecutive triples; $\text{dcsd}(P)$, the minimal distance $|x_i - x_j|$ over *doubly critical vertex pairs*, i.e. pairs where $|x_i - x_j|$ is a local minimum in i and in j (the diagonal $D_{ii} = 0$ is kept in the neighbourhood test, which excludes near-diagonal pairs); and the proxy

$$\tau(P) = \min(\min\text{Rad}(P), \tfrac{1}{2} \text{dcsd}(P)), \quad \text{Rop}(P) = L(P) / \tau(P). \quad (4)$$

The local-minimum test could miss doubly critical pairs of saddle type; as a safeguard we also compute $\tau_{pt}(P) = \min_{i \neq j} |x_j - x_i|^2 / (2 |T_i \times (x_j - x_i)|)$ with central-difference tangents T_i , which uses no criticality test. Neither τ nor τ_{pt} is a certified thickness of a smooth curve (for polygons see [13]); their stability under doubling of N is the main evidence that they approximate the smooth values. Vertex pairs may overestimate the polygon–polygon distance; the minimal segment–segment distance divided by the vertex dcsd is ≥ 0.99998 near the critical pair in every level examined (`experiments/aux_checks.py`). Writhe is the Gauss double sum over pairs of edges (midpoint rule; Limitation (5)).

3 Results

Throughout, $K = K_{d-1}$ is a closed C^∞ regular curve of thickness $\tau = \text{thick}(K) > 0$, hence of maximal curvature $\kappa_{\max} \leq 1/\tau$, carrying the parametrisation of period 2π inherited from Definition 2.1, with speed $v = |K'| \in [v_{\min}, v_{\max}]$; K_d is built by (3) with $r = r_d$, $p = p_d$, $q = q_d$, and $f = r/\tau$.

Lemma 3.1 (embedding). *[proved here] If $r < \tau$ then K_d is a regular C^∞ closed curve, injective on $[0, 2\pi p)$, contained in the torus $\partial\{x : \text{dist}(x, K) < r\}$.*

Proof. Smoothness: the Bishop frame of a C^∞ curve is C^∞ (it solves a linear ODE with smooth coefficients), and (3) is a smooth expression in it; periodicity of K_d in t with period $2\pi p$ holds because $\theta_d(t + 2\pi p) = \theta_d(t) + 2\pi q$ and the closed frame has period 2π . Regularity: differentiating (3) with (2),

$$K'_d = |K'| (1 - r(\kappa_1 \cos \theta + \kappa_2 \sin \theta)) T + r(\theta' + \omega')(-\sin \theta N + \cos \theta B), \quad (5)$$

and $|\kappa_1 \cos \theta + \kappa_2 \sin \theta| \leq \kappa \leq 1/\tau$, so the tangential coefficient is at least $|K'|(1 - r/\tau) > 0$. Injectivity: let $t_1 \neq t_2$ in $[0, 2\pi p)$. If $t_1 \not\equiv t_2 \pmod{2\pi}$, the two points lie in the normal discs of radius $r < \tau$ at distinct points of K ; by definition of thickness these discs are disjoint, so the points differ. If $t_1 \equiv t_2 \pmod{2\pi}$, then $t_1 - t_2 = 2\pi k$ with $0 < |k| < p$ and $\theta_d(t_1) - \theta_d(t_2) = 2\pi qk/p \notin 2\pi\mathbb{Z}$ because $\gcd(p, q) = 1$; the two points lie in the same normal disc at the same radius $r > 0$ but at different angles, hence differ. Containment in the torus is immediate from (3). $\square$

The estimates below use $r \leq \tau/2$; embeddedness only needs $r < \tau$.

Lemma 3.2 (the framing used, and the knot type). *[proved here, from classical results] Let $\alpha \in (-\pi, \pi]$ be the holonomy of the Bishop frame of K . Then the closed frame (N, B) is a framing of K with linking number $n := \text{Lk} = \text{Wr } K - \alpha/2\pi \in \mathbb{Z}$; in particular $\alpha \equiv 2\pi \text{Wr } K \pmod{2\pi}$, and $n = \text{round}(\text{Wr } K)$ whenever $\text{Wr } K \notin \mathbb{Z} + \frac{1}{2}$ (if $\text{Wr } K \in \mathbb{Z} + \frac{1}{2}$ then $\alpha = \pi$ and $n = \text{Wr } K - \frac{1}{2}$). With the convention that a (p, q') -cable of K winds p times along K and q' times around it relative to a framing of linking number 0 (the Seifert framing), K_d is the $(p, q + pn)$ -cable of K [10]. In particular K_1 is the torus knot $T(p_1, q_1)$.*

Proof. By the Călugăreanu–White–Fuller formula [4, 6, 14], $\text{Lk} = \text{Tw} + \text{Wr}$ for any framing. The Bishop frame has zero twist density; the closing rotation ω adds total twist $\text{Tw} = \frac{1}{2\pi} \int \omega' dt = -\alpha/2\pi \in [-\frac{1}{2}, \frac{1}{2}]$. Hence $\text{Lk} = \text{Wr} - \alpha/2\pi$, and since $\text{Lk} \in \mathbb{Z}$ the congruence and the rounding rule follow. The pattern (3) winds q times around the meridian relative to the frame (N, B) while going p times along K ; relative to a framing of linking number 0 it winds $q + pn$ times, which is the stated cable. For K_0 a planar circle, $\text{Wr} = 0 = \alpha$ and K_1 is the standard torus knot. $\square$

Lemma 3.3 (length, two-sided). *[proved here] If $r \leq \tau/2$ then*

$$pL(K)(1 - r\kappa_{\max}) \leq L(K_d) \leq pL(K)(1 + r\kappa_{\max}) + r(2\pi q + p|\alpha|) \leq pL(K)(1 + r/\tau) + r(2\pi q + \pi p). \quad (6)$$

Proof. From (5) and the triangle inequality, $|K'_d| \leq |K'| (1 + r\kappa) + r(|\theta'| + |\omega'|) = |K'| (1 + r\kappa) + r(q/p + |\alpha|/2\pi)$. Integrate over $[0, 2\pi p)$: the first term gives $p \int_0^{2\pi} |K'| (1 + r\kappa) dt \leq pL(K)(1 + r\kappa_{\max})$, the second $r(2\pi q + p|\alpha|)$. Then use $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$. For the lower bound, $|K'_d|$ is at least its tangential component $|K'| (1 - r\kappa) \geq |K'| (1 - r\kappa_{\max})$; integrate. $\square$

Remark 3.4. The term $p|\alpha|r \leq \pi pr$ from the closing twist, absent from a naive count “ p copies plus q meridians”, is needed in the proof because the Bishop frame of a non-planar K does not close (numerically the bound without it also holds in all 36 levels).

Proposition 3.5 (upper bound on the thickness and a lower bound on the ropelength, $p = 2$). *[proved here] Let $p = 2$ (q odd) and $r < \tau$. Then $\text{thick}(K_d) \leq r$, hence $\text{Rop}(K_d) \geq L(K_d)/r \geq pL(K)(1 - r\kappa_{\max})/r$. In the family with $r_d = f \text{thick}(K_{d-1})$ and $f \leq 1/2$ this gives $\text{Rop}(K_d) \geq \frac{p(1-f)}{f} \text{Rop}(K_{d-1})$, so $\text{Rop}(K_d) \geq 2\pi(p(1-f)/f)^d$.*

Proof. The two points $K_d(t)$ and $K_d(t + 2\pi)$ lie in the same normal disc of K at antipodal angles (θ and $\theta + \pi q$, q odd), so their chord is $-2r(\cos \theta N + \sin \theta B)$, orthogonal to T and to $-\sin \theta N + \cos \theta B$, hence by (5) orthogonal to K'_d at both ends: the pair is doubly critical at distance $2r$, so $\text{dcscd}(K_d) \leq 2r$ and $\text{thick}(K_d) \leq r$ by (1) (equivalently, $\rho_{pt} = r$ for this pair). The rest is Lemma 3.3 with $\kappa_{\max} \leq 1/\tau$, $r = f\tau$ and $\text{Rop}(K_{d-1}) = L(K_{d-1})/\tau$. $\square$

At $f = 1/2$ the internal lower bound $2\pi \cdot 2^d$ is 12.6, 25.1, 50.3 at $d = 1, 2, 3$, against the measured 38.4, 155.6, 622.7; $L(K_d)/r_d$ is attained whenever $\text{thick}(K_d) = r_d$ (Section 4). The lower bound of (6) holds in all 36 measured levels.

Thickness of K_d : what is proved and what is not. Write $s_p = 2r \sin(\pi/p)$ for the chord between adjacent strands in a normal disc of K (the p points $K_d(t + 2\pi k)$, $k = 0, \dots, p-1$, sit at the p equally spaced angles $\theta + 2\pi qk/p$, since $\gcd(p, q) = 1$).

Proposition 3.6 (two of the three cases). *[proved here] If $r \leq \tau/2$: (a) two points of K_d in the same normal disc of K are at distance $\geq s_p$; (b) two points of K_d whose base points on K are at distance $\geq 2\tau$ are at distance $\geq 2(\tau - r)$.*

Proof. (a) They are at the same radius r and at angles differing by $2\pi qk/p$, $0 < k < p$, so their distance is $2r|\sin(\pi qk/p)| \geq 2r \sin(\pi/p) = s_p$ because $qk \not\equiv 0 \pmod{p}$. (b) Triangle inequality: $|K_d(t_1) - K_d(t_2)| \geq |K(t_1) - K(t_2)| - 2r$. $\square$

Not covered by Proposition 3.6 are the pairs with $0 < |K(t_1) - K(t_2)| < 2\tau$, and the curvature of K_d . These include nearby base points (helices on a bent tube) and base points on different arcs of K : at $f = 1/2$, $d = 1$ the pair that realises dcscd crosses the hole of the torus (base points 103.7° apart on K_0 , `aux_checks.py`), while for $f \leq 0.35$ it is the antipodal pair of one normal disc. For a straight tube the antipodal pair of the $p = 2$ helices of pitch parameter $h = |K'|p/q$ is a local minimum of the strand distance iff $h \geq r$; on the inner equator of K_1 , $h = (1 - r)p/q \geq r$ iff $r \leq 0.4$ for $(2, 3)$, consistent with $\text{thick}(K_1) = r_1$ for $f \leq 0.35$ only.

Notation for the local case. Let (T, N_0, B_0) be the (non-periodic) Bishop frame of K and α its holonomy, and set

$$m := \frac{q}{p} - \frac{\alpha}{2\pi}, \quad h_{\min} := \frac{v_{\min}}{m}, \quad \Theta_{\min} := \frac{h_{\min}}{\tau}, \quad \zeta_{\min} := \frac{(1 - f)h_{\min}}{r}, \quad f = \frac{r}{\tau} \leq \frac{1}{2}.$$

In the parallel frame the strand of (3) turns at the constant rate m (its angle is $\psi = \theta + \omega = mt + \varphi$), so $h = v/m$ is the pitch parameter above ($|K'|p/q$ when $\alpha = 0$) and $\Theta_{\min}$ is the smallest pitch in units of the base thickness. For two points $X_i = K(t_i) + rU_i$ of K_d with distinct base points, let σ be the length of the *shorter* base arc between $K(t_1)$ and $K(t_2)$, $\delta = \sigma/\tau$, s the base-parameter difference along that arc and $u = m|s|$ the relative turning of the strands along it. The integer $k = (t_2 - t_1 - s)/2\pi \pmod{p}$ is the strand index of X_2 relative to X_1 *along that arc* (for $p = 2$ the two strands are exchanged by one turn around K , so “same strand” is meaningful only relative to a base arc). With $\ell_0(\delta) = 2\tau \sin(\delta/2)$, $E(\delta) = \tau(1 - \cos \delta)$ for $\delta \leq \pi/2$ and $E(\delta) = \tau(1 + \delta - \pi/2)$ for $\delta > \pi/2$, $\beta(\delta) = \delta^2/(4 \sin(\delta/2))$, define

$$\Lambda(\delta) = \ell_0 - \frac{2rE}{\ell_0}, \quad P_1(\delta) = \left[2 \cos\left(\frac{\delta}{2\Theta_{\min}}\right) \sqrt{1 - \beta(\delta)^2} - \frac{\delta^2}{3} \right]_+ \quad (7)$$

(with $P_1 := 0$ if $\delta > \pi\Theta_{\min}$ or $\beta > 1$), and

$$c_1(f, \Theta_{\min}) := \frac{1}{2r} \inf_{0 < \delta < \pi} \max \left\{ \sqrt{\Lambda(\delta)_+^2 + r^2 P_1(\delta)^2}, \ell_0(\delta) - 2r \right\}. \quad (8)$$

All lengths in (8) scale with r , so c_1 depends only on $(f, \Theta_{\min})$; $\ell_0/r - 2$ and $\Lambda/r = (2/f) \sin(\delta/2) - (E/\tau)/\sin(\delta/2)$ decrease in f and P_1 does not depend on f and increases in $\Theta_{\min}$, so c_1 is non-increasing in f and non-decreasing in $\Theta_{\min}$.

Lemma 3.7 (pairs on different strands, $p = 2$). *[proved here] Let $p = 2$, q odd, $r \leq \tau/2$. Any two points X_1, X_2 of K_d with distinct base points and strand index $k = 1$ relative to the shorter base arc satisfy*

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c_1(f, \Theta_{\min})r).$$

Moreover $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$ (a grid evaluation gives 0.6117), hence $c_1 \geq \frac{1}{2}$ whenever $f \leq \frac{1}{2}$ and $\Theta_{\min} \geq \frac{2}{3}$. For the default family $(p, q) = (2, 3)$ with $r_j \leq \text{thick}(K_{j-1})/2$ for all $j \leq d$ one has $\Theta_{\min} \geq \frac{2}{3}$ at every depth, so every such pair is at distance $\geq \min(2(\tau - r), r)$.

Proof. Step 1 (chord of a thick curve). Let $x_0 = K(0)$ and $g(\sigma) = |K(\sigma) - x_0|$ in arclength. A local minimum y_0 of g in $(0, L)$ has $(y_0 - x_0) \perp T(y_0)$, hence $\rho_{pt}(y_0, x_0) = |x_0 - y_0|/2 \geq \tau$, so $|x_0 - y_0| \geq 2\tau$. Consequently $\{g < 2\tau\}$ consists of an initial interval $[0, a)$ and a final one $(L - b, L)$ on which g is monotone (a non-monotone stretch below 2τ would contain an interior local minimum below 2τ), and on them $g' = \cos \angle(T, K - x_0) \geq \sqrt{1 - g^2/4\tau^2}$ a.e. (from $\rho_{pt}(K(\sigma), x_0) \geq \tau$), which integrates to $g(\sigma) \geq 2\tau \sin(\sigma/2\tau)$ and shows $a, b \leq \pi\tau$ (this is the classical chord–arc inequality for thick curves, cf.

[3, 7]). Hence for base points with shorter arc $\sigma = \tau\delta \leq L/2$: either $\delta \geq \pi$ and $|x_0 - y_0| \geq 2\tau$ (then Proposition 3.6(b) gives $|X_2 - X_1| \geq 2(\tau - r)$), or $\delta < \pi$ and $\ell := |x_0 - y_0| \geq \ell_0(\delta)$.

Step 2 (elementary estimates, $\delta < \pi$). Let $x_0 = K(t_1)$, $y_0 = K(t_2)$, $e = (y_0 - x_0)/\ell$ and $\bar{T} = \sigma^{-1} \int vT dt$ over the arc, so $e = \bar{T}/|\bar{T}|$, $|\bar{T}| = \ell/\sigma$. Since $|dT/d\sigma| = \kappa \leq 1/\tau$: (E2) $|\bar{T} - T(x_0)| \leq \delta/2$; (E3) for $w \perp T(x_0)$, $|w \cdot e| = |w \cdot \Pi e| \leq |w| |\bar{T} - T(x_0)| \sigma/\ell \leq |w| \beta(\delta)$, where Π is the projection onto the normal plane at x_0 ; (E4) for unit $w \perp T(x_0)$, $|w \cdot (y_0 - x_0)| = |\int v w \cdot \Pi T dt| \leq \int_0^\sigma \sin \min(\sigma'/\tau, \pi/2) d\sigma' = E(\delta)$, and likewise for $w \perp T(y_0)$; (E5) if $\tilde{U}_2$ denotes the vector with the same parallel-frame angle as U_2 but in the normal plane at x_0 , then $U_2 - \tilde{U}_2 = -\int \kappa_\perp vT dt$ with $|\kappa_\perp| \leq \kappa$ (differentiate $\cos \psi_2 N_0 + \sin \psi_2 B_0$ along the arc), so $|U_2 - \tilde{U}_2| \leq \delta$ and its component orthogonal to e is at most $\int \kappa v |T - \bar{T}| dt \leq \int_0^\sigma \frac{\sigma'^2 + (\sigma - \sigma')^2}{2\sigma\tau^2} d\sigma' = \delta^2/3$; (E6) $\tilde{U}_2 - U_1$ lies in the normal plane at x_0 and has length $2|\sin(\Delta\psi/2)|$, where $\Delta\psi = ms + 2\pi qk/p$ is the relative angle of the strands after parallel transport along the arc (the holonomy terms cancel because $N_0(t + 2\pi)$ is $N_0(t)$ rotated by α and m contains $-\alpha/2\pi$), i.e. $|\sin(\Delta\psi/2)| = |\cos(u/2)|$ for $k = 1$ and $p = 2$.

Step 3 (decomposition). Write $X_2 - X_1 = (y_0 - x_0) + r(U_2 - U_1) = A_e e + Q$ with $Q \perp e$. By (E4), $A_e = \ell + r(U_2 - U_1) \cdot e \geq \ell - 2rE/\ell \geq \Lambda(\delta)$, since $\ell \mapsto \ell - 2rE/\ell$ is increasing and $\ell \geq \ell_0$. By (E3), (E5), (E6), $|Q| = r|(U_2 - U_1)_\perp| \geq r[2|\cos(u/2)|\sqrt{1 - \beta^2 - \delta^2/3}]_+$. Finally $u = m|s| \leq m\sigma/v_{\min} = \delta/\Theta_{\min}$ and $|\cos(u/2)|$ is decreasing on $[0, \pi]$, so $|\cos(u/2)| \geq \cos(\delta/2\Theta_{\min})$ when $\delta \leq \pi\Theta_{\min}$; together with the trivial bound $|X_2 - X_1| \geq \ell - 2r \geq \ell_0 - 2r$ this gives $|X_2 - X_1| \geq 2c_1(f, \Theta_{\min})r$ for every pair with $\delta < \pi$.

Step 4 ($c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$). Put $r = 1$, $\tau = 2$, $\Theta_{\min} = 2/3$; we show that the maximum in (8) is ≥ 1 for every $\delta \in (0, \pi)$. For $\delta \in [\pi/3, \pi/2]$, $\Lambda = 2(\tau - r)\sin(\delta/2) = 2\sin(\delta/2) \geq 1$. For $\delta \in [\pi/2, \pi]$, with $\varsigma = \sin(\delta/2) \in [1/\sqrt{2}, 1]$, $\Lambda = 4\varsigma - (1 + 2\arcsin \varsigma - \pi/2)/\varsigma$, and $\Lambda \geq 1$ amounts to $\phi(\varsigma) := 4\varsigma^2 - \varsigma - 1 + \pi/2 - 2\arcsin \varsigma \geq 0$; $\phi' = 8 - 2\varsigma(1 - \varsigma^2)^{-3/2}$ changes sign once on the interval, so ϕ' is increasing then decreasing, and since $\phi'(1/\sqrt{2}) > 0$ and $\phi'(1^-) = -\infty$, ϕ increases then decreases; its minimum is at an endpoint, $\phi(1/\sqrt{2}) = 0.29$, $\phi(1) = 0.43$. For $\delta \in (0, \pi/3)$ we need $4\sin^2(\delta/2) + P_1^2 \geq 1$. On $[0.6, \pi/3]$: $4\sin^2(\delta/2) \geq 4\sin^2(0.3) > 0.349$, $\cos(3\delta/4) \geq \cos(\pi/4) > 0.7071$, $\beta \leq \beta(\pi/3) < 0.5484$ (β is increasing), so $\sqrt{1 - \beta^2} > 0.8362$, and $\delta^2/3 \leq 0.3656$; hence $P_1 > 2 \cdot 0.7071 \cdot 0.8362 - 0.3656 > 0.8169$, $P_1^2 > 0.667$, and the sum exceeds 1.01. On $(0, 0.6]$: $\cos(3\delta/4) \geq \cos(0.45) \geq 0.900$, $\beta \leq \beta(0.6) \leq 0.305$, $\delta^2/3 \leq 0.12$, so $P_1 \geq 2 \cdot 0.900 \cdot 0.952 - 0.12 \geq 1.59$. All inequalities are non-strict where needed: at $\delta = \pi/3$, $\Lambda = 1$ exactly. Hence $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$. The value 0.6117 (attained near $\delta = 1.23$; an independent evaluation during the review agreed to four digits) is a grid infimum with a Lipschitz constant estimated on the grid itself (`experiments/check_Hc.py`), not a certified bound.

Step 5 (the family). $\Theta_{\min} = v_{\min}/(m\tau)$. By Proposition 3.5, $\tau_{d-1} = \text{thick}(K_{d-1}) \leq r_{d-1} \leq \tau_{d-2}/2$, so $\tau_{d-1} \leq 2^{-(d-1)}$. After the reparametrisation $t \mapsto t/p$, (5) gives $v_{d,\min} \geq p(1 - f)v_{d-1,\min} \geq v_{d-1,\min}$ for $p = 2$, $f \leq 1/2$, so $v_{\min} \geq v_0 = 1$ at every depth; and $m \leq q/p + 1/2 = 2$. Hence $\Theta_{\min} \geq 2^{d-2} \geq 1$ for $d \geq 2$, while for $d = 1$ ($\alpha = 0$, $m = 3/2$, $v = \tau = 1$) $\Theta_{\min} = 2/3$. Monotonicity of c_1 concludes. $\square$

Proposition 3.8 (curvature of K_d). [*proved here, under (H₃)*] Assume in addition (H₃): $\sup |v'| \leq M_v$ and $\sup \sqrt{\kappa_1'^2 + \kappa_2'^2} \leq M_\kappa$ for K (derivatives in t , parallel-frame curvatures; $\kappa_1'^2 + \kappa_2'^2$ includes the torsion: it is $\kappa'^2 + v^2\kappa^2\tau_{\text{tors}}^2$ where $\kappa > 0$). Then, with $g(\varsigma) = \varsigma(\varsigma^2 + 2)/(\varsigma^2 + 1)^{3/2}$ and $G(\varsigma_{\min}) = \sup_{\varsigma \geq \varsigma_{\min}} g \leq g(\sqrt{2}) < 1.0887$: (a)

$$\kappa_{K_d} \leq \bar{\kappa}_d := \frac{G(\varsigma_{\min})}{\tau - r} + \frac{1}{r(1 + \varsigma_{\min}^2)} + \frac{rm[M_v(1 + f) + rv_{\max}M_\kappa]}{(v_{\min}^2(1 - f)^2 + r^2m^2)^{3/2}}, \quad \min\text{Rad}(K_d) \geq 1/\bar{\kappa}_d; \quad (9)$$

(b) $\min\text{Rad}(K_d) \geq 1/\hat{\kappa}_d$, with $A = v^2(1 - x)^2$, $B = r^2m^2$, $b_U = v^2(1 - x)x/r - rm^2$ and

$$\hat{\kappa}_d := \sup_{|x| \leq f, v_{\min} \leq v \leq v_{\max}} \frac{[b_U^2(A + B) + (rm(M_v(1 - x) + rvM_\kappa) + v(A + 2B)\sqrt{\tau^{-2} - x^2/r^2})^2]^{1/2}}{(A + B)^{3/2}}.$$

Proof. Let $U = \cos \psi N_0 + \sin \psi B_0$, $W = T \times U$, $\kappa_\perp = \kappa_1 \cos \psi + \kappa_2 \sin \psi$, $\kappa_W = -\kappa_1 \sin \psi + \kappa_2 \cos \psi$ ($\kappa_\perp^2 + \kappa_W^2 = \kappa^2$), $x = r\kappa_\perp \in [-f, f]$. Differentiating (5) with $U' = mW - \kappa_\perp vT$, $W' = -mU - \kappa_W vT$, $T' = v(\kappa_\perp U + \kappa_W W)$:

$$\begin{aligned} K_d' &= c_T T + c_W W, & K_d'' &= aT + b_U U + b_W W, \\ c_T &= v(1 - x), & c_W &= rm, & b_U &= v^2(1 - x)\kappa_\perp - rm^2, & b_W &= v^2(1 - x)\kappa_W, \end{aligned}$$

$a = a_0 - 2rv m \kappa_W$, $a_0 = v'(1 - x) - rv(\kappa_1' \cos \psi + \kappa_2' \sin \psi)$. Since (T, U, W) is orthonormal and right-handed, $|K_d' \times K_d''|^2 = b_U^2 |K_d'|^2 + (c_W a - c_T b_W)^2$. Put $A = v^2(1 - x)^2$, $B = r^2m^2$, so $|K_d'|^2 = A + B$ and $c_W a -$

$c_T b_W = r m a_0 - \kappa_W v(A + 2B)$. By Minkowski, $\kappa_{K_d} \leq \sqrt{v^4(1-x)^2 \kappa_\perp^2 (A+B) + \kappa_W^2 v^2 (A+2B)^2} / (A+B)^{3/2} + (r m^2 (A+B)^{1/2} + r m |a_0|) / (A+B)^{3/2}$. In the first term $A(A+B) \leq (A+2B)^2$ and $\kappa_\perp^2 + \kappa_W^2 = \kappa^2$ give $\leq v \kappa (A+2B) / (A+B)^{3/2}$, which is decreasing in $A \geq v^2(1-f)^2$; substituting $A = v^2(1-f)^2$ and $\zeta = v(1-f)/(rm)$ yields $\kappa g(\zeta)/(1-f) \leq G(\zeta_{\min})/(\tau(1-f))$. The second term is at most $1/(r(1+\zeta_{\min}^2)) + r m [M_v(1+f) + r v_{\max} M_\kappa] / (v_{\min}^2(1-f)^2 + r^2 m^2)^{3/2}$. (b) By the identity above, $\kappa_{K_d}^2 = [b_U^2(A+B) + (r m a_0 - \kappa_W v(A+2B))^2] / (A+B)^3$ with $x = r \kappa_\perp$, $|a_0| \leq M_v(1-x) + r v M_\kappa$ and $|\kappa_W| \leq (\tau^{-2} - x^2/r^2)^{1/2}$; for fixed (x, v) this is non-decreasing in $|a_0|$ and $|\kappa_W|$. $\square$

Same-strand pairs by a turning argument (added in v0.3). For a pair on the same strand ($k = 0$) Step 3 of Lemma 3.7 gives $|Q| \geq r[2|\sin(u/2)|\sqrt{1-\beta^2-\delta^2/3}]_+$, useless as $u \rightarrow 0$, so short doubly critical pairs must be excluded otherwise. The following measures the turning of the tangent of K_d per unit base arclength and, for $d \geq 2$, uses no derivative of the base (it supersedes a v0.2 argument under (H_3)).

Lemma 3.9 (turning exclusion). *[proved here] Let Γ be a C^1 arc from X_1 to X_2 with unit tangent T_Γ , T_i its tangent at X_i , $a(\sigma) = \angle(T_\Gamma(\sigma), T_1)$ and $b(\sigma) = \angle(T_\Gamma(\sigma), T_2)$. If $a + b < \pi$ along Γ , then $(X_2 - X_1) \cdot (T_1 + T_2) > 0$; in particular (X_1, X_2) is not doubly critical. This holds whenever Γ is $C^{1,1}$ and its total turning $\int_\Gamma |T'_\Gamma|$ is $< \pi$.*

Proof. As $a, b \geq 0$ and $a + b < \pi$, $\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2} > 0$, so $(X_2 - X_1) \cdot (T_1 + T_2) = \int_\Gamma T_\Gamma \cdot (T_1 + T_2) d\sigma > 0$, while a doubly critical pair has $(X_2 - X_1) \cdot T_1 = (X_2 - X_1) \cdot T_2 = 0$. The last sentence follows from $a(\sigma) \leq \int_0^\sigma |T'_\Gamma|$ and $b(\sigma) \leq \int_\sigma^\ell |T'_\Gamma|$ (T_Γ is Lipschitz, hence absolutely continuous). $\square$

In the notation of the proof of Proposition 3.8, $K'_d = v(1-x)T + r m W$ with $x = r \kappa_\perp \in [-f, f]$, so the angle $\chi \in (0, \pi/2)$ between K'_d and T has $\tan \chi = r m / (v(1-x))$ and $\chi_{\text{lo}} := \arctan \frac{r m}{v_{\max}(1+f)} \leq \chi \leq \chi_{\text{hi}} := \arctan \frac{r m}{v_{\min}(1-f)} = \arctan \frac{1}{\zeta_{\min}}$.

Proposition 3.10 (pairs on the same strand, $p = 2$, sharpened). *[proved here] Let $p = 2$, q odd, $f = r/\tau \leq \frac{1}{2}$, $m > 0$. Set $\delta_{R2} := (\pi - 2(\chi_{\text{hi}} - \chi_{\text{lo}}))/(1 + \sin \chi_{\text{hi}}/\Theta_{\min})$; if (H_3) of Proposition 3.8 holds, set $\delta_{R1} := \pi/(\tau \bar{\Omega})$ with*

$$\tau \bar{\Omega} := 1 + \sin^2 \chi_{\text{hi}} + \frac{\sin \chi_{\text{hi}}}{\Theta_{\min}} + \tau \frac{r m [M_v(1+f)/v_{\min} + r M_\kappa]}{v_{\min}^2(1-f)^2 + r^2 m^2},$$

and otherwise $\delta_{R1} := 0$; let $\delta_{\text{turn}} = \max(\delta_{R1}, \delta_{R2})$. Then every doubly critical pair of K_d with distinct base points and strand index $k = 0$ relative to the shorter base arc has $\delta \geq \delta_{\text{turn}}$ and

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c'_0 r), \quad c'_0 := \frac{1}{2r} \inf_{\delta_{\text{turn}} \leq \delta < \pi} \max(\Lambda(\delta), \ell_0(\delta) - 2r).$$

In particular $c'_0 \geq \frac{1}{2}$ whenever $\delta_{\text{turn}} \geq \pi/3$.

Proof. If $\delta \geq \pi$ or the base chord is $\geq 2\tau$, Step 1 of Lemma 3.7 and Proposition 3.6(b) give $|X_2 - X_1| \geq 2(\tau - r)$. Otherwise let Γ be the arc of K_d over the shorter base arc, $t \in [t_1, t_2]$, so $u = m(t_2 - t_1) \leq \delta/\Theta_{\min}$; since $k = 0$, Γ joins X_1 to X_2 .

$\delta \geq \delta_{R2}$, without (H_3) . Put $\chi_i = \chi(t_i)$ and $Y_i(t) = \cos \chi_i T(t) + \sin \chi_i W(t)$. From $U' = mW - v \kappa_\perp T$, $W' = -mU - v \kappa_W T$, $T' = v(\kappa_\perp U + \kappa_W W)$ we get $Y'_i = (v \kappa_\perp \cos \chi_i - m \sin \chi_i)U + v \kappa_W \cos \chi_i W - v \kappa_W \sin \chi_i T$ and $|Y'_i|^2 = (v \kappa_\perp \cos \chi_i - m \sin \chi_i)^2 + v^2 \kappa_W^2 \leq (v \kappa + m \sin \chi_i)^2$, using $\kappa_\perp^2 \cos^2 \chi_i + \kappa_W^2 \leq \kappa^2$ and $-\kappa_\perp \cos \chi_i \leq \kappa$. The unit tangent $T_d(t) = \cos \chi(t) T(t) + \sin \chi(t) W(t)$ and $Y_i(t)$ lie in $\text{span}(T(t), W(t))$ at angles $\chi(t)$, χ_i from $T(t)$, and $Y_i(t_i) = T_d(t_i)$, so $a(t) \leq |\chi(t) - \chi_1| + \int_{t_1}^t (v \kappa + m \sin \chi_1)$ and $b(t) \leq |\chi(t) - \chi_2| + \int_t^{t_2} (v \kappa + m \sin \chi_2)$. With $\int v \kappa dt \leq \sigma/\tau = \delta$, $a + b \leq 2(\chi_{\text{hi}} - \chi_{\text{lo}}) + \delta + u \sin \chi_{\text{hi}} \leq 2(\chi_{\text{hi}} - \chi_{\text{lo}}) + \delta(1 + \sin \chi_{\text{hi}}/\Theta_{\min})$, which is $< \pi$ if $\delta < \delta_{R2}$; Lemma 3.9 excludes the pair.

$\delta \geq \delta_{R1}$, under (H_3) . The turning rate of T_d per unit t is $\varpi = |K'_d \times K''_d|/|K'_d|^2$. The Minkowski split in the proof of Proposition 3.8, multiplied by $|K'_d| = \sqrt{A+B}$, gives $\varpi \leq v \kappa (A+2B)/(A+B) + r m^2/\sqrt{A+B} + r m |a_0|/(A+B)$, where $B/(A+B) = \sin^2 \chi$ and $r m/\sqrt{A+B} = \sin \chi$; with $\kappa \leq 1/\tau$, $\chi \leq \chi_{\text{hi}}$, $v \geq v_{\min}$ and $|a_0| \leq M_v(1+f) + r v M_\kappa$ this is $\varpi/v \leq \bar{\Omega}$ per unit base arclength, so the total turning of Γ is $\leq \tau \bar{\Omega} \delta$, which is $< \pi$ if $\delta < \delta_{R1}$.

Distance. For $\delta < \pi$, Step 3 of Lemma 3.7, which does not use the strand index, gives $|X_2 - X_1| \geq \Lambda(\delta)$ and $|X_2 - X_1| \geq \ell_0(\delta) - 2r$; hence c'_0 . Finally $\Lambda \geq r$ on $[\pi/3, \pi)$ when $f \leq \frac{1}{2}$: on $[\pi/3, \pi/2]$, $\Lambda = 2(\tau - r) \sin(\delta/2) \geq \tau - r \geq r$; on $[\pi/2, \pi)$ this is Step 4 of Lemma 3.7, and Λ/r decreases in f . $\square$

Corollary 3.11 (same strand in the default family). *[proved here] Let $(p_j, q_j) = (2, 3)$ and $f_j = r_j/\text{thick}(K_{j-1}) \leq \frac{1}{2}$ for all $j \leq d$. Then every doubly critical pair of K_d with distinct base points on the same strand is at distance $\geq \min(2(\tau - r), 2c'_0 r)$ with $c'_0 \geq \frac{1}{2}$, at every depth $d \geq 1$, without (H_3) . At $d = 1$ and $f = \frac{1}{2}$, $\delta_{\text{turn}} \geq \delta_{R1} = \pi/(1 + \frac{9}{13} + \frac{9}{2\sqrt{13}}) > 1.0684$ and $c'_0 \geq \sin(\delta_{R1}/2) > 0.509$ (and $c'_0 > 0.509$ for every $f \leq \frac{1}{2}$); for $d \geq 2$, $\delta_{\text{turn}} \geq \delta_{R2} > 1.677$.*

Proof. $d = 1$. The base is the unit circle: $v \equiv 1$, $\kappa \equiv 1 = 1/\tau$, $\alpha = 0$, $m = \frac{3}{2}$, $\Theta_{\min} = \frac{2}{3}$, and (H_3) holds with $M_v = M_\kappa = 0$. With $\zeta = \zeta_{\min} = 2(1 - f)/(3f)$, $\tau\bar{\Omega} = 1 + \frac{1}{1+\zeta^2} + \frac{3}{2\sqrt{1+\zeta^2}}$, which is $< 2.9404 < 3$ at $f = \frac{1}{2}$ ($\zeta = \frac{2}{3}$) and decreases as f decreases; so $\delta_{R1}(f) \geq \delta_{R1}(\frac{1}{2}) > \pi/3$. We argue at $f = \frac{1}{2}$, with $c_* := 2\sin(\delta_{R1}(\frac{1}{2})/2)$ fixed. On $[\delta_{R1}(\frac{1}{2}), \pi/2]$, $\Lambda/r = 2(1/f - 1)\sin(\delta/2) \geq c_*$. On $[\pi/2, \pi)$ the argument of Step 4 of Lemma 3.7 with 1 replaced by c_* goes through: $\phi_{c_*}(\zeta) = 4\zeta^2 - c_*\zeta - 1 + \pi/2 - 2\arcsin \zeta$ has the same second derivative, $\phi'_{c_*}(1/\sqrt{2}) = 2\sqrt{2} - c_* > 1.81$, $\phi_{c_*}(1/\sqrt{2}) = 1 - c_*/\sqrt{2} > 0.279$ and $\phi_{c_*}(1) = 3 - c_* - \pi/2 > 0.410$; so $c'_0(\frac{1}{2}) \geq c_*/2 > 0.509$. For $f < \frac{1}{2}$ the interval $[\delta_{\text{turn}}(f), \pi)$ is contained in $[\delta_{R1}(\frac{1}{2}), \pi)$ and Λ/r increases as f decreases, so $c'_0(f) \geq c_*/2$ as well.

$d \geq 2$. By (5), $|K'_1|^2 = (1 - x)^2 + r_1^2 m_1^2$ with $r_1 = f_1$, $m_1 = \frac{3}{2}$, so after the reparametrisation $v_{\min}(K_1) \geq 2\sqrt{(1 - f_1)^2 + \frac{9}{4}f_1^2}$, while $\text{thick}(K_1) \leq f_1$ (Proposition 3.5); hence $v_{\min}(K_1)/\text{thick}(K_1) \geq 2\sqrt{(1 - f_1)^2/f_1^2 + \frac{9}{4}} \geq \sqrt{13}$. For $j \geq 2$, $v_{\min}(K_j) \geq 2(1 - f_j)v_{\min}(K_{j-1})$ and $\text{thick}(K_j) \leq f_j \text{thick}(K_{j-1})$, so $v_{\min}/\text{thick}$ at least doubles per depth. With $m_d \leq \frac{3}{2} + \frac{1}{2} = 2$, $\Theta_{\min} \geq \sqrt{13}2^{d-3} \geq \sqrt{13}/2 > 1.802$ and $\zeta_{\min} = (1 - f_d)\Theta_{\min}/f_d \geq \Theta_{\min}$. Dropping $\chi_{10} \geq 0$, $\delta_{R2} \geq (\pi - 2\arctan(1/\zeta_{\min}))/((1 + 1/(\Theta_{\min}\sqrt{1 + \zeta_{\min}^2})))$, which increases in $\Theta_{\min}$ and $\zeta_{\min}$ and at $\Theta_{\min} = \zeta_{\min} = \sqrt{13}/2$ exceeds $1.677 > \pi/3$. (The bound $\Theta_{\min} \geq 2^{d-2}$ of Step 5 of Lemma 3.7 would not suffice at $d = 2$: at $f = \frac{1}{2}$, $\delta_{R2} \geq \pi/3$ needs $\Theta_{\min} \geq 1.112$.) Proposition 3.10 concludes. $\square$

Theorem 3.12 (thickness lower bound for $p = 2$). *[proved here; (b) under (H_3) ; the values in Table 1 are grid evaluations, not certified] Let $p = 2$, q odd, $f = r/\tau \leq 1/2$, $m > 0$. (a) $\frac{1}{2}\text{dcsd}(K_d) \geq \min(\tau - r, \min(c_1, c'_0)r)$, with $c_1 = c_1(f, \Theta_{\min})$ of Lemma 3.7 and c'_0 of Proposition 3.10. (b) Under (H_3) , $\min\text{Rad}(K_d) \geq 1/\hat{\kappa}_d$, hence $\text{thick}(K_d) \geq \min(\tau - r, cr)$ with $c = \min(c_1, c'_0, 1/(r\hat{\kappa}_d))$. (c) For the default family $(2, 3)$ with $f_j \leq \frac{1}{2}$ at every level, $\frac{1}{2}\text{dcsd}(K_d) \geq \min(\tau - r, r/2) = r/2$ and $\text{thick}(K_d) \geq \min(\min\text{Rad}(K_d), r/2)$ at every depth, without (H_3) ; at $d = 1$, $\text{thick}(K_1) \geq r_1/2$; for $d \geq 2$, (H_c) with $c = \frac{1}{2}$ holds at level d if and only if $\min\text{Rad}(K_d) \geq r_d/2$.*

Proof. By (1), $\text{thick} = \min(\min\text{Rad}, \frac{1}{2}\text{dcsd})$. Doubly critical pairs with the same base point are at distance $2r$ (Proposition 3.6(a)); if the shorter base arc has $\delta \geq \pi$, the base chord is $\geq 2\tau$ (Step 1 of Lemma 3.7) and the distance is $\geq 2(\tau - r)$ (Proposition 3.6(b)); pairs with $\delta < \pi$ are covered by Lemma 3.7 ($k = 1$) and Proposition 3.10 ($k = 0$). Since $c_1 \leq 1$ (its limit as $\delta \rightarrow 0$ is 1), this gives (a); (b) is Proposition 3.8(b). For (c), $c_1 \geq \frac{1}{2}$ is Lemma 3.7, $c'_0 \geq \frac{1}{2}$ is Corollary 3.11, and $\tau - r \geq r$. At $d = 1$ ($M_v = M_\kappa = 0$, $v \equiv 1$), Proposition 3.8 gives $r\hat{\kappa}_1 = fG(\zeta_{\min})/(1 - f) + 1/(1 + \zeta_{\min}^2)$ with $\zeta_{\min} = 2(1 - f)/(3f)$ and $G \leq \max g = g(\sqrt{2}) = 4\sqrt{2}/(3\sqrt{3})$ (the derivative of $y(y + 2)^2/(y + 1)^3$, $y = \zeta^2$, is proportional to $(y + 2)(2 - y)$); this increases with f , and at $f = \frac{1}{2}$ equals $g(\sqrt{2}) + \frac{9}{13}$, so $1/(r\hat{\kappa}_1) > 0.561 > \frac{1}{2}$. The equivalence for $d \geq 2$ follows from $\text{thick} = \min(\min\text{Rad}, \frac{1}{2}\text{dcsd})$, $\frac{1}{2}\text{dcsd} \geq r/2$ and $\min(\tau - r, \frac{r}{2}) = \frac{r}{2}$. $\square$

In every measured level $c \geq 0.509$ (Table 1; $f = 0.25, 0.35$ in `results/theory_constants.md`).

Table 1: Constants of Theorem 3.12 on the polygons at $f = \frac{1}{2}$ ($N_0 = 256$; τ, v, α and, for $1/(r\hat{\kappa}_d)$ at $d \geq 2$, M_v, M_κ measured on the base polygon; $\delta_{\text{turn}} = \delta_{R1}$ at $d = 1$, δ_{R2} at $d \geq 2$), rounded down; last column: measured τ_d/r_d . Grid evaluations, not certified (`experiments/check_Hc.py`, `check_same_strand.py`); analytic: $c_1, c'_0 \geq \frac{1}{2}$. At $d = 2$ the column $1/(r\hat{\kappa}_d)$ is superseded by Proposition 3.19.

f	d	c_1	c'_0 (δ_{turn})	$1/(r\hat{\kappa}_d)$	c	τ_d/r_d
0.5	1	0.611	0.509 (1.068)	1.419	0.509	0.832
0.5	2	0.699	0.845 (2.056)	1.034	0.699	1.000
0.5	3	0.706	0.988 (2.928)	0.998	0.706	1.000

Remark 3.13 (no constant works for an arbitrary base). *[proved here] No $c > 0$ depending only on (p, q, f) makes $\text{thick}(K_d) \geq \min(\tau - r, cr \sin(\pi/p))$ true for every smooth base K with $\text{thick}(K) = \tau$. (i) If the curvature of K rises from 0 to $1/\tau$ over an arc of length ε (a smoothed junction of a segment and a circle of*

radius τ), by (5) the tangent of the inner strand turns from $vT + rmW$ to $v(1-f)T + rmW$ over that arc, so $\kappa_{K_d} \gtrsim rv\kappa'/|K'_d|^2 \rightarrow \infty$: the M_v, M_κ term of (9) is unavoidable. (ii) If K is traversed slowly ($v \rightarrow 0$) on an arc of length $\gg r$, the two strands there are helices of pitch $h = v/m \rightarrow 0$; in the straight-tube model $|K_d(t) - K_d(t + 2\pi + u)|^2 = 4r^2 \cos^2(qu/4) + v^2u^2$, whose minimum over u is $\leq (2\pi v/q)^2$ and is attained at a doubly critical pair by screw symmetry, so $\text{thick}(K_d) \leq \pi v/q \rightarrow 0$ (in Lemma 3.7, $c_1 \rightarrow 0$ as $\Theta_{\min} \rightarrow 0$). In the family $\Theta_{\min}$ is bounded below a priori (Lemma 3.7, Corollary 3.11); no a priori bound on M_v, M_κ uniform in d is known (but see Corollary 3.16 and, for $d = 2$, Proposition 3.19).

Towards (H_3) uniformly in d (added in v0.4). For a regular closed curve K with parameter t of period 2π , speed v , curvature vector $\kappa = dT/d\sigma = \kappa_1 N_0 + \kappa_2 B_0$ and $\kappa'_{\text{par}} := \kappa'_1 N_0 + \kappa'_2 B_0$ (the normal part of $d\kappa/dt$), set $\nu(K) := \sup_t |v'|/v^2$ and $\mu(K) := \sup_t |\kappa'_{\text{par}}|/v$. ν is invariant under linear reparametrisations ($\tilde{v}(s) = pv(ps)$ gives $\tilde{v}'/\tilde{v}^2 = v'/v^2$), $\mu = \sup |\Pi_\perp d\kappa/d\sigma|$ is geometric, and $M_v \leq \nu v_{\max}^2$, $M_\kappa \leq \mu v_{\max}$. In the notation of the proof of Proposition 3.8 put $\varepsilon := r\kappa_{\max}(K) (\leq f$ if $r = f\tau)$, $\lambda := rm/v_{\min}$ and $\zeta' := v_{\min}(1-\varepsilon)/(rm) (\geq \zeta_{\min}$ if $r = f\tau)$. Lemma 3.14 and Proposition 3.15 hold for every $r > 0$ with $\varepsilon < 1$ and $m > 0$: their proofs use neither τ nor $f \leq \frac{1}{2}$. The four statements below were derived in `theory/` after the second review, checked by the author (symbolic frame calculus and a smooth computation, `experiments/verify_H3.py`) and by an independent internal referee (round 4).

Lemma 3.14 (speed of K_d). *[proved here; checked by an independent internal referee (round 4)]* With $w = |K'_d|$, $c_T = v(1-x)$ and $a_0 = v'(1-x) - rv\kappa'_{\text{par}} \cdot U$,

$$w' = \frac{c_T(a_0 - rvm\kappa_W)}{w}, \quad \text{hence} \quad \nu(K_d) \leq \frac{\nu(K)}{1-\varepsilon} + \frac{r\mu(K) + \lambda\kappa_{\max}(K)}{(1-\varepsilon)^2}. \quad (10)$$

Proof. $w^2 = c_T^2 + r^2m^2$ with rm constant, so $w' = c_T c'_T/w$, and $c'_T = v'(1-x) - vx'$ with $x' = r(\kappa'_{\text{par}} \cdot U + m\kappa_W)$ (as $\kappa \cdot U' = m\kappa_W$). Then $|w'|/w^2 \leq |a_0 - rvm\kappa_W|/c_T^2$ (since $w \geq c_T$), and $|v'| \leq \nu v^2$, $|\kappa'_{\text{par}}| \leq \mu v$, $|\kappa_W| \leq \kappa_{\max}$, $1-x \geq 1-\varepsilon$, $v \geq v_{\min}$ give the bound; $\nu(K_d)$ may be computed in any linear reparametrisation. $\square$

Proposition 3.15 (curvature of K_d from the data of the base). *[proved here; checked by an independent internal referee (round 4)]* For every $r > 0$ with $\varepsilon = r\kappa_{\max}(K) < 1$ (no condition on τ or f), with G as in Proposition 3.8,

$$\kappa_{\max}(K_d) \leq \frac{\kappa_{\max}(K)G(\zeta')}{1-\varepsilon} + \frac{1}{r(1+\zeta'^2)} + \frac{\lambda(\nu(K)(1+\varepsilon) + r\mu(K))}{(1-\varepsilon)^3}. \quad (11)$$

Proof. The Minkowski split of the proof of Proposition 3.8(a), which is algebraic, with $|x| \leq \varepsilon < 1$ and $\kappa \leq \kappa_{\max}(K)$: the first term is $\leq v\kappa(A+2B)/(A+B)^{3/2}$, decreasing in $A \geq v^2(1-\varepsilon)^2$, hence $\leq \kappa_{\max}G(\zeta')/(1-\varepsilon)$; $rm^2/(A+B) \leq 1/(r(1+\zeta'^2))$; and $rm|a_0|/(A+B)^{3/2} \leq rm v^2(\nu(1+\varepsilon) + r\mu)/(v^3(1-\varepsilon)^3) \leq \lambda(\nu(1+\varepsilon) + r\mu)/(1-\varepsilon)^3$. $\square$

Corollary 3.16 (tolerated growth of the (H_3) data, default family). *[proved here; checked by an independent internal referee (round 4)]* Let $(p_j, q_j) = (2, 3)$ and $r_j \leq f \text{thick}(K_{j-1})$, $f \leq \frac{1}{2}$, for all $j \leq d$, and let $d \geq 3$. If

$$\frac{96}{\sqrt{13}} 4^{-d} \nu(K_{d-1}) + \frac{64}{\sqrt{13}} 8^{-d} \mu(K_{d-1}) \leq 2 - g(\sqrt{13}) - \frac{1}{14} \quad (= 0.8961\dots), \quad (12)$$

then $\min\text{Rad}(K_d) \geq r_d/2$, hence $\text{thick}(K_d) \geq r_d/2$: (13) below holds at depth d with $c = \frac{1}{2}$. (Each term may take the whole right-hand side only if the other vanishes: $\nu(K_{d-1}) \leq 0.0336 \cdot 4^d$ with $\mu = 0$, or $\mu(K_{d-1}) \leq 0.0504 \cdot 8^d$ with $\nu = 0$; in general the joint condition (12) applies.)

Proof. By the proof of Corollary 3.11, $\zeta' \geq \zeta_{\min} \geq \Theta_{\min} \geq \sqrt{13}2^{d-3} \geq \sqrt{13}$, so $G(\zeta') \leq g(\sqrt{13})$ (g decreases on $[\sqrt{2}, \infty)$) and $1/(1+\zeta'^2) \leq 1/14$; $\varepsilon \leq \frac{1}{2}$ gives $\varepsilon/(1-\varepsilon) \leq 1$, $(1+\varepsilon)/(1-\varepsilon)^3 \leq 12$, $(1-\varepsilon)^{-3} \leq 8$; Proposition 3.5 gives $r_d \leq 2^{-d}$; and $\lambda = (1-f_d)/\zeta_{\min} \leq 2^{3-d}/\sqrt{13}$. Multiplying (11) by r_d , $r_d\kappa_{\max}(K_d) \leq g(\sqrt{13}) + \frac{1}{14} + 12r_d\lambda\nu + 8r_d^2\lambda\mu \leq 2$ under (12); Theorem 3.12(c) concludes. $\square$

Proposition 3.17 (the recursion for (H_3) does not close). *[proved here; checked by an independent internal referee (round 4)]* Fourth derivatives of K enter $K_d'''' = K'''' + rU''''$ only through the term $-rv(\kappa''_{\text{par}} \cdot U)T$ of rU'''' : with $n_\chi := \sin \chi T - \cos \chi W$ the unit normal of K_d in $\text{span}(T, W)$ and $\sin \chi = rm/w > 0$, the normal part of K_d'''' is $-rv \sin \chi (\kappa''_{\text{par}} \cdot U) n_\chi + R$, with $|R|$ bounded in terms of $v_{\max}, \kappa_{\max}, \nu, \mu, \sup |v''|, r, m$. Consequently no locally bounded F satisfies $\mu(K_d) \leq F(v_{\min}, v_{\max}, \kappa_{\max}, \nu, \mu, \sup |v''|, r, m)$ for all smooth bases K : the speed and curvature of K_d are controlled by the (H_3) data of K , but $\mu(K_d)$ is not.

Proof. $U''' = mW'' - (v\kappa_\perp)''T - 2(v\kappa_\perp)'T' - v\kappa_\perp T''$, where $W'' = -mU' - (v\kappa_W)'T - v\kappa_W T'$ and $T'' = v'\kappa + v d\kappa/dt$ involve at most third derivatives of K , and $\kappa_\perp'' = \kappa_{\text{par}}'' \cdot U + 2m\kappa_{\text{par}}' \cdot W - m^2\kappa_\perp$; the normal part of T is $\sin \chi n_\chi$ (a symbolic computation in the frame (T, U, W) confirms the coefficient; v'' enters R with coefficient $(1-x)\sin \chi$). For the last claim take a smooth closed K with $r < \text{thick}(K)$ and $K_\eta = K + \eta^{7/2}\Phi((t-t_0)/\eta)U(t_0)$ with Φ smooth, compactly supported, $\Phi''''(0) \neq 0$. For small η the support lies in one period, K_η is regular and $r < \text{thick}(K_\eta)$, so $K_{d,\eta}$ is defined and embedded (thickness is lower semicontinuous under C^2 convergence: minRad is continuous, and doubly critical pairs at distance $\leq 2r$ converge, after extraction, to a doubly critical pair of K , collapsing pairs being excluded by Lemma 3.9). $K_\eta \rightarrow K$ in C^3 , so the arguments of F converge, while $|\kappa_{\text{par}}'' \cdot U|(t_0) \gtrsim \eta^{-1/2}$; since the normal part of K_d'''' is $3ww'\kappa_d + w^3\Pi_\perp d\kappa_d/d\sigma$ with w, w', κ_{K_d} bounded by Lemma 3.14 and Proposition 3.15, $\mu(K_{d,\eta}) \rightarrow \infty$. $\square$

Remark 3.18 (what this gives and what it does not). [grid evaluation with polygon inputs; uniformity open] Lemma 3.14 and Proposition 3.15 propagate ν and κ_{max} given μ at the previous depth, and Corollary 3.16 reduces (H_c) with $c = \frac{1}{2}$ at depth $d \geq 3$ to a growth condition. This does *not* prove (H_c) for $d \geq 3$: by Proposition 3.17 an a priori bound on $\mu(K_{d-1})$ cannot be obtained from the (H_3) data of K_{d-2} alone; a recursive argument must carry higher derivatives (e.g. all of them, via analyticity) or use the explicit structure of the family. Proposition 3.17 says nothing about the family itself, where μ grows slowly (values below). At $d = 2$ the a priori constants do not suffice (the analogue of (12), $\frac{6}{\sqrt{13}}\nu(K_1) + \frac{1}{\sqrt{13}}\mu(K_1) \leq 2 - g(\frac{\sqrt{13}}{2}) - \frac{4}{17} < 0.685$, fails: the polygons give 1.95); $d = 2$ is settled by Proposition 3.19. On the polygons ($f = \frac{1}{2}$) μ goes from 4.5 at $d = 1$ to 6.9 at $d = 4$ (smooth: 7.0), the left-hand side of (12) is ≤ 0.48 ($d = 3$) and ≤ 0.15 ($d = 4$), and (10), (11) hold in all 16 checked levels with bound/measured ≥ 1.01 (experiments/check_H3.py, smooth_H3.py; all values in results/theory_constants.md).

The case $d = 2$ (added in v0.5). At $d = 2$ the base K_1 is an explicit trigonometric polynomial, so instead of the a priori constants one can insert into (11) rigorous enclosures of the four numbers of K_1 that it uses.

Proposition 3.19 (the case $d = 2$). [computer-assisted proof (interval arithmetic); checked by an independent internal referee (round 5)] Let $(p_1, q_1) = (p_2, q_2) = (2, 3)$, $0 < r_1 \leq \frac{1}{2}$ and $0 < r_2 \leq r_1/2$, with arbitrary phases φ_1, φ_2 (equivalently, arbitrary initial normals of the frames of K_0 and K_1). Then

$$r_2 \kappa_{\text{max}}(K_2) \leq 1.187, \quad \text{hence} \quad \text{minRad}(K_2) \geq 0.842 r_2 > r_2/2.$$

Consequently, for every $f \leq \frac{1}{2}$, every $r_1 \leq f$ and every $r_2 \leq f$ $\text{thick}(K_1)$, $\text{thick}(K_2) \geq r_2/2$: Hypothesis 3.21 with $c = \frac{1}{2}$ holds at $d = 2$ for every choice of phases. Along the family $r_1 = f$, $r_2 \leq f^2$ one has moreover $\text{minRad}(K_2) > r_2$ for every $f \leq 0.484375$.

Proof. Reduction. By Proposition 3.15 with $K = K_1$, $r = r_2$, which needs only $\varepsilon = r_2 \kappa_{\text{max}}(K_1) < 1$ and not $f \leq \frac{1}{2}$ ($r_2/\text{thick}(K_1)$ may exceed $\frac{1}{2}$: at $r_1 = \frac{1}{2}$ the polygonal $\tau_1 = 0.4158 < 2r_2$; the script checks $\varepsilon \leq 0.355$ on every box), $r_2 \kappa_{\text{max}}(K_2) \leq \Phi$, where Φ is r_2 times the right-hand side of (11), a function of $\kappa_{\text{max}}(K_1)$, $v_{\text{min}}(K_1)$, $\nu(K_1)$, $\mu(K_1)$, r_2 and m_2 ; its proof uses them only through $|\kappa| \leq \kappa_{\text{max}}$, $|v'| \leq \nu v^2$, $|\kappa'_{\text{par}}| \leq \mu v$, $v \geq v_{\text{min}}$, so bounds may be inserted. Φ is non-decreasing in κ_{max} , ν , μ , r_2 and m_2 and non-increasing in v_{min} : $\varepsilon = r_2 \kappa_{\text{max}}$ and $\lambda = r_2 m_2 / v_{\text{min}}$ increase, $\zeta' = v_{\text{min}}(1 - \varepsilon)/(r_2 m_2)$ decreases, and G is non-increasing. Since $\alpha_2 \in (-\pi, \pi]$, $m_2 = \frac{3}{2} - \alpha_2/2\pi < 2$, so m_2 may be replaced by 2: the Bishop frame of K_1 does not enter. This covers both sides of $r_1 = r_1^* \approx 0.4904$, where α_2 crosses π , m_2 jumps from ≈ 1 to ≈ 2 and the cable slope of K_2 jumps (Lemma 3.2, Section 4). By Proposition 3.5, $\text{thick}(K_1) \leq r_1$, so $r_2 \leq f \text{thick}(K_1)$ with $f \leq \frac{1}{2}$ implies $r_2 \leq r_1/2$. A change of φ_1 moves K_1 by a rigid motion and a shift of the parameter ($K_1^\varphi(s) = R_z(-2\varphi/3)K_1^0(s + \varphi/3)$), which leave the four numbers unchanged; φ_2 enters (11) only through $|U| = 1$.

The base. With $t = 2s$, $K_1(s) = (e^{2is}(1 + r_1 \sin 3s), r_1 \cos 3s)$ (first two coordinates as a complex number; $x + iy = e^{2is} + \frac{r_1}{2i}(e^{5is} - e^{-is})$), with explicit derivatives $D_j = d^j K_1/ds^j$. With $v = |D_1|$: $\kappa = |D_1 \times D_2|/v^3$, $|v'|/v^2 = |D_1 \cdot D_2|/v^3$ and, from $\Pi_\perp K_1''' = 3vv'\kappa + v^2\Pi_\perp \kappa'$ and $\kappa = \Pi_\perp D_2/v^2$, $|\kappa'_{\text{par}}|/v = |\Pi_\perp D_3 - 3(D_1 \cdot D_2)\Pi_\perp D_2/v^2|/v^3$, where $\Pi_\perp$ is the projection orthogonal to D_1 . Moreover $K_1(s + 2\pi/3) = R_z(4\pi/3)K_1(s)$ exactly, so suprema over $[0, 2\pi]$ equal suprema over $[0, 2\pi/3]$.

Enclosure. Cover $[0, \frac{1}{2}] \times [0, 2\pi/3]$ by 512×2048 boxes in (r_1, s) (r_1 -endpoints exact binary fractions, s -endpoints enclosed in `mpmath.iv`); on each box enclose $\cos ks, \sin ks$ ($k = 1, 2, 3, 5$) with `mpmath.iv` (80 bits, rounded outward to binary64), and evaluate the expressions above in binary64 interval arithmetic

with every endpoint rounded outward by one ulp after each IEEE-754 operation ($+$, $-$, $\times$, $\div$, $\sqrt{}$), which encloses the exact value; the script stops if an endpoint is not finite or $\varepsilon \geq 1$. On each r_1 -box $[a, b]$ this gives upper bounds of $\kappa_{\max}$, ν , μ and a lower bound of $v_{\min}$ over the box; Φ evaluated in the same arithmetic at these values with $r_2 = b/2$ and $m_2 = 2$ is at most 1.187 on every box (the maximum is on the last box, $r_1 \in [0.499, 0.5]$). Theorem 3.12(c) gives $\text{thick}(K_2) \geq r_2/2$. For the last claim the same box data with $r_2 = b^2$ give $\Phi < 1$ on every box with $b \leq 0.484375$. $\square$

Remark 3.20 (what is certified, and how). The certificate covers exactly $r_2\kappa_{\max}(K_2) \leq 1.187$ for $r_1 \leq \frac{1}{2}$, $r_2 \leq r_1/2$; the rest is Theorem 3.12(c), proved by hand. The trust base is IEEE-754 binary64 arithmetic in NumPy in round-to-nearest mode without flush-to-zero (the defaults), Python's `float()` of an `mpmath` number rounding to nearest, `mpmath.iv`, and about 150 lines of code (`experiments/certify_d2.py`, the script of `theory/` with assertions and comments added in v0.6; 8s); it is not a formal verification. The implementation was checked by the author (other formulas, 368 640 point-in-box tests; `experiments/check_certify_d2.py`) and by an independent internal referee (round 5: line-by-line audit and a separate enclosure); see the README. The printed constant is that of this interval extension. The bound is not sharp: (11) takes the four suprema separately and splits by Minkowski (at $r_1 = \frac{1}{2}$ the actual m_2 is 1.98, so $m_2 \leq 2$ costs little). Smooth (uncertified) values of $r_2\kappa_{\max}(K_2)$, $r_2 = r_1/2$, are ≈ 0.40 at $r_1 = \frac{1}{2}$ and default phases, ≈ 0.41 at most over the phases there, and ≈ 0.48 at most over the sampled phases and $r_1 \leq \frac{1}{2}$, just below r_1^* (`verify_H3.py`, `smooth_d2_phases.py`), against 1.187; so $\text{minRad}(K_2) > r_2$ is certified only for $f \leq 0.484375$. Nothing at $d \geq 3$, and no polygonal quantity, is certified.

What remains unproved is therefore a statement about the family, not about an arbitrary curve:

Hypothesis 3.21 (H_c , for the family). [for (2, 3) and $c = 1/2$: proved at $d = 1$ and (computer-assisted) at $d = 2$; for $d \geq 3$ reduced to a curvature bound (Theorem 3.12) and to a growth condition (Corollary 3.16); verified numerically] Let (K_d) be the family of Definition 2.1 built from the circle with a fixed pattern (p, q) , and let $f \leq 1/2$. There is $c = c(p, q, f) \in (0, 1]$ such that, for every sequence (r_d) with $r_d \leq f \text{thick}(K_{d-1})$ and every $d \geq 1$,

$$\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, c r_d \sin(\pi/p)). \quad (13)$$

Status for (2, 3), $f \leq 1/2$, $c = 1/2$. By Theorem 3.12(c), (13) holds at $d = 1$ and, for $d \geq 2$, is equivalent to $\text{minRad}(K_d) \geq r_d/2$. At $d = 2$ this is Proposition 3.19 (computer-assisted; every $f \leq \frac{1}{2}$ and every choice of phases). For $d \geq 3$ it follows from $1/(r\hat{\kappa}_d) \geq 1/2$ under (H_3) (0.998 at $d = 3$, Table 1) and from (12), which the polygons satisfy at $d = 3, 4$ (Remark 3.18). This is evidence, not a proof: ν , μ (and, for $\hat{\kappa}_d$, $\text{thick}(K_{d-1})$) are only measured. Open: (H_3) uniformly in d (here, a priori control of $\mu(K_{d-1})$) and certification for $d \geq 3$.

On the constant. The natural constant is $c = 1$: $r \sin(\pi/p)$ is half the strand chord, and by Propositions 3.5 and 3.6(a) it cannot be improved where $\tau_d = r_d$. With $c = 1$ the inequality fails on the polygons in 4 of the 15 measured levels with $f \leq 1/2$ ((2, 3), $f = 0.5$, $d = 1$ (0.832); (3, 2), $d = 1$ (0.968); (3, 2), $d = 2$ (0.999); (2, 5), $d = 1$ (0.564); the deficit at (3, 2), $d = 2$ is stable under doubling of N) and holds with equality, to 0.05%, where $\tau_d = r_d$ and in (3, 2), $d = 3$; (2, 5), $d = 2, 3$. The value $c = 1/2$ was chosen after $c = 1$ had failed; Theorem 3.12 reaches it, while $\tau_1/r_1 = 0.832$ at $f = 1/2$ shows that no argument reaches $c = 1$ there.

Corollary 3.22 (recursion for the ropelength bound). [proved here; for (2, 3), $c = \frac{1}{2}$, $f \leq \frac{1}{2}$: unconditional for $d \leq 2$ (Proposition 3.19 computer-assisted), conditional on $\text{minRad}(K_d) \geq r_d/2$ for $d \geq 3$] Fix (p, q) , $f \leq 1/2$ and assume (H_c) with constant c (for all $r_d \leq f \text{thick}(K_{d-1})$). Define $\rho_0 = 1$, $r_d := f\rho_{d-1}$ and $\rho_d := \gamma\rho_{d-1}$ with

$$\gamma = \min(1 - f, c f \sin(\pi/p)), \quad \mathcal{A} = p(1 + f) + f(q + \frac{p}{2}). \quad (14)$$

Then $\text{thick}(K_d) \geq \rho_d = \gamma^d$ and $L(K_d) \leq 2\pi\mathcal{A}^d$ for all d , hence $\text{Rop}(K_d) \leq 2\pi\Lambda_{\text{Rop}}^d$ with $\Lambda_{\text{Rop}} = \mathcal{A}/\gamma$. For (2, 3) and $f = 1/2$: $\mathcal{A} = 5.00$, and $\Lambda_{\text{Rop}} = 20.0$ with $c = 1/2$ ($\gamma = 0.250$), $\Lambda_{\text{Rop}} = 10.0$ with $c = 1$ ($\gamma = 0.500$).

Proof. Induction on d . If $\text{thick}(K_{d-1}) \geq \rho_{d-1}$ then $r_d = f\rho_{d-1} \leq f \text{thick}(K_{d-1})$, so (H_c) applies and $\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, c r_d \sin(\pi/p)) \geq \min(\rho_{d-1} - f\rho_{d-1}, c f \rho_{d-1} \sin(\pi/p)) = \gamma\rho_{d-1} = \rho_d$. By Lemma 3.3 with $\tau \geq \rho_{d-1}$, $L(K_d) \leq pL(K_{d-1})(1 + f) + f\rho_{d-1}(2\pi q + \pi p)$; since $\rho_{d-1} \leq \text{thick}(K_{d-1}) \leq L(K_{d-1})/2\pi$, this is $\leq L(K_{d-1})[p(1 + f) + f(q + p/2)] = \mathcal{A}L(K_{d-1})$, and $L(K_0) = 2\pi$. For (2, 3), $c = \frac{1}{2}$, $f \leq \frac{1}{2}$, the hypothesis is proved at $d = 1$ (Theorem 3.12(c)) and at $d = 2$, where $r_2 = f\rho_1 \leq f \text{thick}(K_1)$ (Proposition 3.19). $\square$

Table 2: Chains at the finest resolution ($N_0 = 1024$; 512 for $(3, 2)$). $r_d = f\tau_{d-1}$; $\rho_d^{(1)}$ is (13) with $c = 1$, so $\tau_d/\rho_d^{(1)} < 1$ marks a failure of $c = 1$; slope = $q + pn$ (Lemma 3.2). $f = 0.25$ (as $f = 0.35$) and the probe $f = 0.6$ (outside $r \leq \tau/2$): `results/tables.md`.

(p, q)	f	d	N	r_d	L	τ	L/τ	$\tau_d/\rho_d^{(1)}$	Wr	slope
(2, 3)	0.35	1	2048	0.3500	14.271	0.3500	40.8	1.000	3.26	3
(2, 3)	0.35	2	4096	0.1225	28.608	0.1225	233.5	1.000	15.49	9
(2, 3)	0.35	3	8192	0.0429	57.219	0.0429	1334.6	1.000	63.98	33
(2, 3)	0.5	1	2048	0.5000	15.949	0.4158	38.4	0.832	3.52	3
(2, 3)	0.5	2	4096	0.2079	32.349	0.2079	155.6	1.000	17.99	11
(2, 3)	0.5	3	8192	0.1040	64.730	0.1040	622.7	1.000	74.96	39
(3, 2)	0.5	1	1536	0.5000	20.000	0.4191	47.7	0.968	3.90	2
(3, 2)	0.5	2	4608	0.2096	60.087	0.1813	331.4	0.999	39.66	14
(3, 2)	0.5	3	13824	0.0906	180.270	0.0785	2296.5	1.000	362.97	122
(2, 5)	0.5	1	2048	0.5000	20.420	0.2822	72.4	0.564	6.87	5
(2, 5)	0.5	2	4096	0.1411	41.116	0.1411	291.4	1.000	32.66	19
(2, 5)	0.5	3	8192	0.0705	82.273	0.0705	1166.2	1.000	136.30	71

Remark 3.23 (on the constants). Λ_{Rop} is generous ($c = 1/2$ is twice weaker than the observed $\tau_d \approx r_d$): along the chain of the corollary ($r_d = f\rho_{d-1}$, `experiments/corollary_chain.py`) the polygons give $L/\tau = 38.4, 256.5, 2051.9$ against $2\pi\Lambda_{\text{Rop}}^d = 126, 2513, 50265$, and $\tau_d/\rho_d \geq 1.66$ for $f \in \{0.25, 0.35, 0.5\}$. No configuration is claimed stationary or optimal, and the recursion is not a “flow law” for tight recursive knots: it bounds one explicit construction.

Remark 3.24 (lower bounds, quoted). [literature] Buck and Simon [2] prove $\text{Rop}(K) \geq c_{\text{BS}} \text{Cr}(K)^{3/4}$ for a universal $c_{\text{BS}} > 0$; Cantarella, Kusner and Sullivan [3] prove that ropelength minimisers exist in every knot type, are $C^{1,1}$ and satisfy further lower bounds. These need $\text{Cr}(K_d)$. K_2 is a 2-cable of the trefoil (Lemma 3.2), which is adequate; by Kalfagianni and McConkey [8] a cable with q strands of an adequate knot K has crossing number $\geq q^2 \text{Cr}(K) + 1$, so $\text{Cr}(K_2) \geq 13$ (their exact value $4 \text{Cr}(K) + 1$ covers the slopes $\pm 5, \pm 7$ of the trefoil, not the slopes 9, 11 of K_2). For $d \geq 3$ only general satellite bounds are available, such as $\text{Cr} \geq 10^{-13} \text{Cr}(\text{companion})$ [9]. For $K_1 = T(2, 3)$ the best known numerical ropelength is about 32.7 (uncertified; [3, 12]) against our 38.4 at $f = 1/2$: the construction is not close to tight even at depth one.

Conjecture 3.25 (a threshold for $\text{thick}(K_d) = r_d$). [conjectural] For the pattern $(2, 3)$ there are thresholds $f_*(d) > 0$ with $\inf_d f_*(d) > 0$ such that for $f \leq f_*(d)$ and every choice of the phases $\varphi_1, \dots, \varphi_d$ (equivalently, of the frame normalisations) every doubly critical pair of K_d other than the antipodal pairs of one normal disc is at distance $> 2r_d$ and $\min\text{Rad}(K_d) > r_d$; hence $\text{thick}(K_d) = r_d$ (the inequality $\leq$ is Proposition 3.5). At $d = 2$ the part $\min\text{Rad}(K_2) > r_2$ is proved for $f \leq 0.484375$ (Proposition 3.19). Evidence, at the grid values $f \in \{0.25, 0.35, 0.5\}$ only: $f_*(1) \in [0.35, 0.5)$, and $\tau_d = r_d$ at $d = 2, 3$ for $f \leq \frac{1}{2}$, at $f = 0.25, 0.35$ for the default phases and at $f = \frac{1}{2}$ for every phase tested, with polygonal margins $\geq 1.37\%$ ($d = 2$) and $\geq 0.233\%$ ($d = 3$, joint scan), smooth values being slightly smaller (Section 4).

If $\text{thick}(K_d) = r_d$ for all large d and $\min\text{Rad}(K_{d-1})$ stays bounded below, then $\text{Rop}_d / \text{Rop}_{d-1} \rightarrow p/f$ by Lemma 3.3; the measured ratios (4.06, 4.00 at $f = 1/2$ against $p/f = 4.00$) agree: the rate is a consequence of the conjecture, not an independent constant.

4 Experiments

Criteria. The protocol is in Appendix A. Four criteria are checked: (C1) resolution: Rop and τ change by less than 1% between N_0 and $2N_0$; (C2) $L(K_d) \leq$ the right-hand side of (6) evaluated with the measured $L(K_{d-1})$, $\tau(K_{d-1})$ and α_d ; (C3) $\tau(K_d) \geq \rho_d$ of (13) with $c = 1/2$ and the measured $\tau(K_{d-1})$, for $f \leq 1/2$; (C4) τ_{pt} agrees with τ to within 1%. The plan written before any code (`PLAN.md`) fixed the resolution check, the length bound and the form of (13); the 1% thresholds, (C1) at $d = 3$ and (C4) were fixed when the manuscript was written, and $c = 1/2$ in (C3) after a run with $c = 1$ had failed at ($f = 1/2, d = 1$): (C3) is a post hoc check.

Results. Table 2 gives all patterns at the finest resolution (plots of the growth and of the curves: `figures/fig_rop.pdf`, `fig_curves.pdf`). All four criteria hold: (C1) the maximal relative change is

0.03% for Rop and 0.03% for τ over all 12 chains (per-level values in `results/tables.md`); (C2) holds in 36 of 36 levels; (C3) holds in all 9 default levels with $f \leq 1/2$, with margin $\tau_d/\rho_d \geq 1.66$ (≥ 1.13 with the other patterns); (C4) the maximal discrepancy is 0.01%. In every level τ is determined by $\text{dcsc}/2$, never by minRad (36 of 36; $\text{minRad}/r_d \geq 1.15$ throughout).

Three further observations. First, for $f \leq 0.35$ the measured $\tau(K_d)$ equals r_d to four digits at every depth (the antipodal strands of a normal disc are the doubly critical pair; Proposition 3.5, Conjecture 3.25), so $\text{Rop}_d = L_d/r_d$; for $f = \frac{1}{2}$, $\tau_1 = 0.4158 < r_1$, realised by the pair across the hole of the torus (Section 3), while at $d = 2, 3$ again $\tau_d = r_d$, but narrowly: the next doubly critical vertex pair is only 1.4% and 2.1% farther than $2r_d$. These margins depend on the phases (Definition 2.1): over φ_2 the $d = 2$ margin is $\geq 1.37\%$, the default being at the minimum (`experiments/phase_margin.py`), and a joint scan of (φ_2, φ_3) at $d = 3$ gives $\geq 0.233\%$, never negative, with $\tau_d = r_d$ throughout (`joint_phase_sweep.py`). These are grid evaluations on vertex pairs, biased upwards; a smooth construction (`smooth_margin.py`) gives 1.369% at $d = 2$ and 0.214% at the minimiser of the joint scan at $d = 3$ (scan sizes, periods and refinements: README). The census of `classify_pairs.py` finds same-strand doubly critical pairs with base arc $< \pi\tau$ only at $d = 1$, with $\delta \geq 1.96 > \delta_{\text{turn}}$. Second, $\text{Wr}(K_1)$ crosses 3.5 inside the grid (3.257 at $f = 0.35$, 3.518 at $f = 0.5$; crossing at $r_1^* = 0.4904$ by bisection), so by Lemma 3.2 the frame linking number is 3 for $f \leq 0.35$ and 4 for $f \geq 0.5$: K_2 is the $(2, 9)$ -cable of the trefoil for $f = 0.25$ but the $(2, 11)$ -cable for $f = 0.5$, robustly to the writhe error (Limitation (5), `results/convergence.md`). Third, $f = 0.5$ gives the best value on the grid at depths one and three, $f = 0.6$ is worse at every depth (`results/tables.md`); f was not optimised.

5 Claims and their status

Claim	Status
K_d embedded; K_d the $(p, q + pn)$ -cable; length bounds (Lemmas 3.1–3.3).	proved here (Lemma 3.2 from classical results)
$p = 2$: $\text{thick}(K_d) \leq r_d$, $\text{Rop}(K_d) \geq 2\pi(p(1-f)/f)^d$ (Prop. 3.5). Distances of same-disc, far, different-strand and same-strand pairs; $c_1, c'_0 \geq \frac{1}{2}$ in $(2, 3)$, all d , no (H_3) (Prop. 3.6, Lemma 3.7, Prop. 3.10, Cor. 3.11).	proved here (internal to the family) proved here; refined values (0.6117, Table 1) are grid values
$\text{minRad}(K_d) \geq 1/\bar{\kappa}_d, \geq 1/\hat{\kappa}_d$ (Prop. 3.8).	proved under (H_3) ; unconditional at $d = 1$; at $d = 2$ superseded by Prop. 3.19
$(2, 3)$, $f \leq \frac{1}{2}$: $\frac{1}{2} \text{dcsc}(K_d) \geq r_d/2$ for all d ; $\text{thick}(K_1) \geq r_1/2$; for $d \geq 2$, $(H_c) \Leftrightarrow \text{minRad}(K_d) \geq r_d/2$ (Thm. 3.12). $\nu, \kappa_{\max}$ of K_d from the data of K_{d-1} ; (H_c) for $d \geq 3$ under (12); $\mu(K_d)$ not bounded by the (H_3) data of an arbitrary base (Lemma 3.14–Prop. 3.17).	proved here proved here
$\text{minRad}(K_2) \geq 0.842r_2$, so $\text{thick}(K_2) \geq r_2/2$, for every $f \leq \frac{1}{2}$ and all phases; $\text{minRad}(K_2) > r_2$ for $f \leq 0.484375$ (Prop. 3.19).	computer-assisted proof (interval arithmetic); checked by an independent internal referee (round 5)
(12) at $d = 3, 4$; (10), (11) on the polygons; $\text{thick}(K_3) \geq r_3/2$ (Remark 3.18, Table 1). No $c > 0$ depending only on (p, q, f) works for every base (Remark 3.13).	grid evaluation, polygon inputs proved here
(H_c) , $(2, 3)$, $c = \frac{1}{2}$, $d \geq 3$; (H_3) uniformly in d . $\text{Rop}(K_d) \leq 2\pi\Lambda_{\text{Rop}}^d$, $\Lambda_{\text{Rop}} = \mathcal{A}/\gamma$ (Cor. 3.22).	open; numerically true in 9 levels, margin ≥ 1.66 proved; unconditional for $d \leq 2$ ($(2, 3)$, $c = \frac{1}{2}$), conditional on (H_c) for $d \geq 3$
Polygons with $L/\tau = 38.4, 155.6, 622.7$ ($f = 0.5$, $d = 1, 2, 3$), stable to 0.03%.	computed (polygonal proxy, uncertified)
K_2 changes knot type between $f = 0.35$ and 0.5 ($r_1^* = 0.4904$). $\text{Rop} \geq c_{\text{BS}} \text{Cr}^{3/4}$; tight knots; $\text{Cr}(K_2) \geq 13$; satellites.	computed; consequence of Lemma 3.2 literature [2, 3, 8, 9]
$\text{thick}(K_d) = r_d$ for $f \leq f_*(d)$ and all phases; then $\text{Rop}_d / \text{Rop}_{d-1} \rightarrow p/f$ (Conj. 3.25).	conjectural; narrow margins at $f = \frac{1}{2}$ (Section 4)
Stationarity, optimality, a recursive flow law.	not claimed

6 Limitations and next steps

Limitations. (1) The thickness lower bound ($p = 2$, default family) is proved for all doubly critical pairs at every depth, completely at $d = 1$ and, computer-assisted, at $d = 2$ (Proposition 3.19; not sharp; checked by an independent internal referee, round 5); for $d \geq 3$ the curvature of K_d needs (H_3) , whose constants are only measured, and their propagation in d is partial (Corollary 3.16, Proposition 3.17). (2) The

polygonal τ of (4) is supported by τ_{pt} , the segment check and stability in N , but is not certified. (3) Nothing is optimised; the family is one of many; the margins of Conjecture 3.25 depend on the phases, scanned only at $f = \frac{1}{2}$; for $p \geq 3$ the lower bound is treated only in Proposition 3.6. (4) Besides the analytic thresholds ($c_1, c'_0 \geq \frac{1}{2}$, $1/(r\bar{\kappa}_1) > 0.561$, Corollaries 3.11, 3.16) and Proposition 3.19, every constant of Section 3 is a grid value with polygon-measured inputs for $d \geq 2$. (5) Writhe and holonomy are midpoint-rule sums, stable to $1 \cdot 10^{-4}$ at depth one but only to 0.031 at depth three of (3, 2); the frame linking number for $f = 0.5$ is decided by a margin of 0.018. (6) The only family-specific lower bound is Proposition 3.5; knot-type bounds need $\text{Cr}(K_d)$ (Remark 3.24).

Next steps. (1) Make (H_3) uniform in d : by Proposition 3.17 this needs control of all derivatives, e.g. by induction on analyticity in a complex strip; with Corollary 3.16 this would prove (H_c) , $c = 1/2$, for (2, 3) at every depth, and Corollary 3.22 unconditionally. At $d = 3$, $\nu(K_2)$, $\mu(K_2)$ depend only on the local data of K_1 (up to fourth derivatives), r_2 , m_2 and the angle of U : an interval computation over boxes in (s, angle, r_2) would settle $d = 3$. (2) Certify c_1 , c'_0 , $\hat{\kappa}_d$ ($d \geq 3$) and minRad , dcsd of a piecewise-circular interpolant; exact polygonal writhe. (3) A constrained length minimisation at fixed τ with a certified knot-type check ($d = 1, 2$; the probe `experiments/shrink.py` gave nothing usable). (4) Optimise f_d and the pattern angle; treat $p \geq 3$. (5) Match the conventions of [8] with Lemma 3.2 to obtain $\text{Cr}(K_2)$ and a Buck–Simon bound per configuration.

A Reproducibility and protocol

Protocol. `experiments/recursive_knots.py` builds the polygon K_0 with N_0 vertices, computes the closed Bishop frame by discrete parallel transport (Rodrigues rotations; initial normal as in Definition 2.1; holonomy removed by the twist $-\alpha i/N$) and produces K_d with $p_d N_{d-1}$ vertices by (3) at the vertices (t is the vertex index). At each depth it measures L , minRad , dcsd , τ , τ_{pt} , Rop , writhe, holonomy and $\text{Lk} = \text{rint}(\text{Wr } K_{d-1} - \alpha_d/2\pi)$ with its residual, and sets $r_{d+1} = f\tau(K_d)$. Grid: (2, 3) with $f \in \{0.25, 0.35, 0.5, 0.6\}$, (3, 2), (2, 5) with $f = 0.5$; depths 1–3; $N_0 \in \{512, 1024\}$ ($\{256, 512\}$ for (3, 2)). Deterministic; 184s wall-clock, Python 3.11.15, NumPy 2.4.6. The auxiliary scripts (constants of Section 3, census, chain of Corollary 3.22, phase scans, smooth computations, the computer-assisted proof of Proposition 3.19 and its checks; SHA-256 of the frozen ones in `results/FROZEN_THEORY_SHA256.txt`) and their run times are listed in the README. `make_numbers.py` writes `numbers.tex` and the table body from `results/*.json` only. Every number in this text is a macro from `numbers.tex`.

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